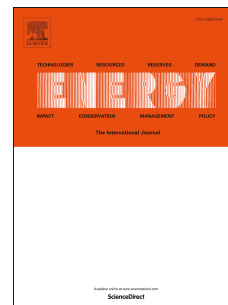


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First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming

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Abstract

Thermochemical recuperation of waste flue gas heat may be advantageous for improving energy efficiency of hydrocarbon fuel-consuming furnace. The schematic diagram of thermochemical recuperation (TCR) by steam methane reforming is described. Thermodynamics equilibrium analysis of steam methane reforming (SMR) process has been investigated via Gibbs free energy minimization technique to determine the effects of pressure, inlet steam-to-methane ratio and temperature on TCR efficiency. The energy analysis was carried out for temperature range and steam-to-methane ratio of 600-1300 K and 1-6, respectively, at different pressure of 1-20 bar. The results shown that TCR efficiency can be controlled for maximum energy efficiency by the operating pressure, temperature and various inlet feed stocks. The recuperation rate and heat balance of TCR were analyzed for different waste flue gas temperatures, steam-to-methane ratio at mixture inlet and pressure. For the effective operation of thermochemical recuperation it is necessary to use a pressure of less than 10 bar, because in this case the percentage by volume of non-combustible components in the synthesis gas is minimal. The optimal operation conditions for TCR were determined: steam-to-methane ratio is 2 for flue gas temperature 900-1100K; steam-to-methane ratio is 1 for the temperature range above 1200K; optimal pressure is 5-10bar.

Keywords: Steam methane reforming, Hydrogen, Energy efficiency,

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Thermochemical recuperation, Waste-heat, Synthesis gas

Nomenclature

β	steam-to-methane ratio;
λ	excess air ratio;
μ_i	chemical potential of element i ;
G^t	value of total Gibbs free energy;
G_{fi}^0	standard Gibbs free energy of formation of element i ;
R	molar gas constant;
T	temperature;
f_i	fugacity of pure elements i ;
f_i^0	fugacity of pure elements i at the standard state;
X_{CH_4}	equilibrium conversion rates of methane;
$[CH_4]_{in(out)}$	inlet (outlet) mole fraction of methane;
H_{SMR}	enthalpy of steam methane reforming reaction;
H_{sg}	enthalpy of combustible components of synthesis gas;
H_{fg}	enthalpy of hot flue gases after combustion chamber;
$H_{synfuel}^{out}$	total enthalpy of syn.gas flow at outlet temperature;
C_{p_i}	specific heat of i component;
H_{steam}	enthalpy of steam after steam generator;
D	mass flow of steam;
h_g	specific enthalpy of saturated steam;
LHV	low heating value;
SMR	steam methane reforming;
TCR	thermochemical recuperation.

1. Introduction

Natural gas is a very important raw hydrocarbon in the social and industrial development of all countries in the world. At the present moment, the global fuel

5 consumption has been sharply increasing, and the energy resources supply has become the most significant value for the governments in all countries [1]. The high level of fossil fuel consumption is the reason of ecological and environmental concerns which is implementing the civilized society today [2–4]. The natural gas consumption in Russia from 2006 to 2016 is approximately 450-460 billion cubic
 10 meters per year [5, 6]. About 11% of all natural gas in Russia is consumed in heat technology: industrial gas furnaces, high-temperature plants, etc [5]. Also, the energy efficiency of such plants is very low compared with same plants in the developed countries. It is reported [5, 7], the energy saving potential in Russia is about 40% of the current level. Today, the average energy efficiency
 15 of heat technology in Russia is only about 31%, which is tens of percent lower than that of the industrial developed countries [5, 8]. For example, according to the report of the Russian Ministry of Energy, in Russia the average energy efficiency of steelmaking furnaces is about 31%, forging furnaces is about 46%, glass-melting furnaces is about 45% [7]. The main reason for the low energy
 20 efficiency of such plants is the large heat loss of waste flue gases [9–12].

Today, the using of heat of waste flue gas appears to be very important in order to achieve better energy efficiency of industrial furnaces. Improving energy efficiency has a focus on manufacturers' Research & Development efforts for last years and significant progress has been made [13–15] . One of the
 25 main ways to increase the energy efficiency of industrial furnaces is the waste-heat recovery. Recovering a part of these heat losses can give a significant improvement of energy efficiency and as a result decreasing of fuel combustion in those furnaces. Considering industrial furnaces, it can be argued that the greatest heat loss occurs due to flue gases. For example, heat losses with flue
 30 gases after steelmaking furnaces can reach 65-75%, in forges about 55-60%, glass furnaces about 50-60% [16–19]. Here, the heat losses with flue gases are defined as relation between the heat of exhaust gases to the low heating value of prime fuel [16]. The recovery of flue gas waste-heat is very important for efficiency industrial furnace exploitation due to production of significant volume of high
 35 temperature exhaust gas. Engineering solutions for industrial furnaces rely on

air preheating using recuperators and regenerators after combustion chamber to improve energy efficiency [20–22].

In recent years, researchers and engineers are paying more attention to novel ways of waste-heat recovery. In this case could be mentioned works on the renewable approaches to recovering wasted heat by thermoelectrics [23–25] and thermionics [26] waste heat recuperation systems. These methods of heat recuperation are direct conversion from waste heat to electricity.

Another relatively perspective way of heat recovery of hot flue gas streams is using the waste-heat to partially reform of the natural gas. In other words the recuperating of furnace's waste heat is possible by using heat of flue gases to carry out endothermic reactions of initial fuel reforming that produce hydrogen and carbon monoxide-rich synthesis gas [27–30]. This new synthetic fuel has greater low heating value than original gas fuel (for example, natural gas) and can be burnt more efficiently in the combustion chamber in comparison to the initial fuel [31, 32]. Since, the natural gas reforming is strongly endothermic it gives an opportunity for recuperation of waste flue gas energy in a thermochemical form that otherwise would have been wasted. This method usually called as thermochemical recuperation (TCR) [33–35]. TCR has been receiving innovated interest as one of the possible methods of improving energy efficiency of industrial furnaces. Thermochemical waste-heat recuperation can be carried out by various reforming reactions: steam methane reforming [30, 33, 36], steam reforming of ethanol [29, 35], reforming of isooctane [28, 37], steam-dry methane reforming [33], methanol steam reforming [38], diesel reforming [39], etc [40]. Results about the advantages of thermochemical recuperation vary extensively because many various schematic diagrams have been investigated.

Thermochemical waste-heat recuperation can be used in various fuel-consumption plants. Verkhivker and Kravchenko [30] reported that thermochemical waste-heat recuperation is the most considerable way of improving of energy efficiency of combined-cycle plants. They calculated that the total energy efficiency of a combined cycle plant with thermochemical waste-heat recuperation has obtained about 80–90%. It is possible because heat utilization in TCR is not limited by

isentropic expansion (Carnot efficiency). In addition, there are articles that show the use of dry-methane reforming for increasing the energy efficiency of gas-turbine cycles [41, 42]. The study of Cocco and coauthors [43] reports a
 70 significant gain in first law efficiency of 13% with thermochemical recuperation by usage of dimethyl ether.

Popov S.K et al. [33] reported that thermochemical recuperation of flue gases heat can be a promising way to improving energy efficiency of high-temperature plants which use natural gas, for example glass melting furnaces. They showed
 75 that TCR can reach saving up to 10-25% of energy without affecting other parameters of the process. In that investigation the options of glass melting units with thermochemical waste flue gas recovery in the recuperating reactor were studied. Also, Maruoka N. et al. [44] investigated the steel-making furnace with thermochemical waste-heat recuperation by steam methane reforming.

80 Kobayashi [37] patented the thermochemical regenerative heat recovery process for industrial furnaces. A furnace is provided having at least two regenerator beds for heat recovery. While a first bed is being heated by hot flue gases produced by fuel combustion in the furnace, a second bed, heated during a previous cycle, is cooled through carrying out an endothermic chemical reforming reaction
 85 (for example, steam methane or dry methane reforming). Once the second bed is cooled by the endothermic reaction, the hot flue gases are redirected to it while the first bed, now hot, is used for carrying out the endothermic chemical reaction. Thereafter the cycle is repeated. Barati and co-authors [18] studied the method of utilization of thermal energy of the metallurgical slag by thermochemical recuperation. Also, thermochemical waste-heat recuperation via
 90 fuel reforming can be considered as on-board hydrogen production technology [28, 38, 45, 46].

Rue D. and co-workers [34] calculated the energy efficiency increasing of glass furnace with thermochemical recuperation. They reported that they have created the model of thermochemical waste-heat recuperation and tested it. The
 95 result of their work is the fact that energy efficiency of furnace with TCR is significantly higher than it is with thermal recuperators or other exchangers.

Besides, an investigation of glass furnaces with TCR with steam methane reforming have been published in other articles [17, 47–49]. In these papers the process conditions (temperature, pressure, inlet feed ratio) for syngas production have been studied. It was reported that complete conversion of methane can be achieved for temperatures above 700 °C at inlet feed steam/methane ratio of about 3.0.

In all the articles mentioned above, it is pointed out that the main advantage of the TCR is high energy efficiency. All of these papers focus on the increase in the low heating value (LHV) of reforming product (syngas) compared to the primary fuel [50]. Also, Knoche and et al. [31, 32] presented that the combustion irreversibility of synthesis gas is less than is of the primary fuel. In other words, thermochemical recuperation of waste flue gases heat is a perspective way to decrease fuel consumption for various furnaces and other fuel-consuming plants. Therefore, further study of thermochemical recuperation is topical.

Research studies on thermochemical waste-heat recuperation by steam methane reforming mentioned above were mainly focused on investigation of various schematic diagrams and thermodynamic analysis of original fuel reforming. However; there is no literature (according to the author's literary search) on energy analysis results for overall thermochemical recuperation process in terms of further effects of flue gas temperature, pressure and feed composition on overall energy efficiency of TCR process, in particular, recuperation rate. For example, Popov et al. [33] suggested to use inlet gas mixture with composition of $\text{CH}_4:\text{H}_2\text{O}$ as 1:2 for TCR by steam methane reforming. Maruoka and coauthors [44] analyzed of heat flows of the steelmaking furnace with thermochemical recuperator for $\text{CH}_4:\text{H}_2\text{O} = 1:1$. But in this paper it was assumed that the conversion rate of methane is 1, the effect of the inlet mixture composition and temperature has not been taken into account. As a rule, other works also do not answer the question about the necessary operation conditions, that should be used for maximum efficiency of thermochemical waste-heat recuperation system by using steam methane reforming. To answer this question, it is necessary to combine the thermodynamic analysis of the steam methane reforming and the

calculation of the heat flows for all the thermochemical waste-heat recuperation
 130 system.

In the present work, energy analysis of thermochemical recuperation system by steam methane reforming was studied. The effects of pressure, various steam-to-methane feed compositions and the flue gas temperature on the TCR efficiency were investigated. The heat balance of furnace with thermochemical
 135 recuperation was compiled to determine the optimal conditions of operating (flue gas temperature, steam-to-methane ratio at inlet reformer and pressure). The maximum recuperation rate of the thermochemical heat recuperation system is determined for different operating conditions. To determine enthalpy of steam methane reforming reaction and synthesis gas composition the thermodynamic
 140 analysis was accomplished by means of total Gibbs free energy minimization via IVTANTHERMO. Simulation of heat balance of TCR system was performed using Mathcad for wide range of flue gas temperature 600-1300 K.

2. Thermochemical recuperation concept

2.1. Schematic diagram

145 The main concept of thermochemical waste-heat recuperation is transformation of flue gases heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low heating value. Fig. 1 illustrates the application of the thermochemical recuperation principle in natural gas-consuming furnace. In this schematic diagram, the hot waste flue gas after the combustion
 150 chamber of the gas furnace are consistently passed through reformer and steam generator (boiler). In the reformer, heat is indirectly extracted from the hot waste flue gas to reform some or all of the prime fuel to synthesis gas. In the reactor volume of the reformer, the steam methane reforming occurs to produce synthesis gas [51–53]. The synthesis gas after reformer contains varying mix-
 155 tures of the raw fuel, hydrogen, carbon monoxide, steam, carbon dioxide and others. After reformer the syngas (new synthetic fuel) is fed to the combustion chamber of gas furnace for combustion in air stream. Inter-cooled waste

flue gas after reformer is passed through steam generator for steam producing.
Steam and methane are mixed in steam ejector and then the resulting reaction
mixture is fed to reaction space of reformer.

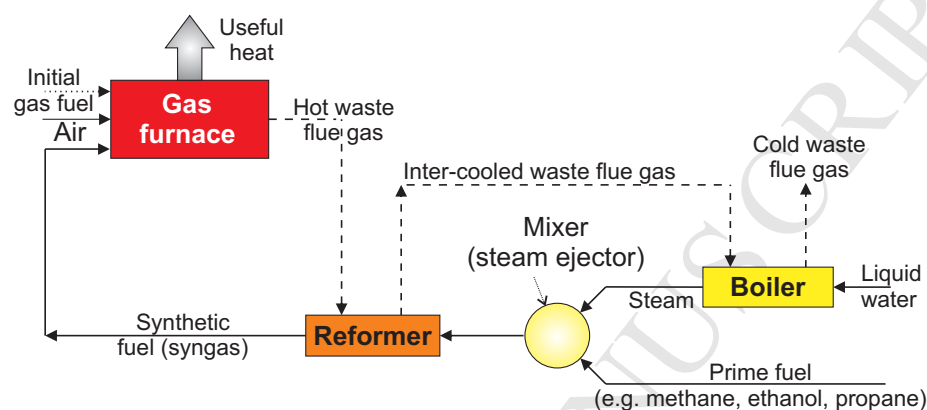


Figure 1: The schematic diagram of natural gas-consuming furnace with thermochemical recuperation of flue gases

Fig. 2 illustrates two major types of reformers for thermochemical recuperation of flue gas waste heat: tubular (a) and plate (b). Numerical study for the design of a reactor for steam methane reforming has been conducted with respect to the variable operating and design parameters [54]. For improving efficiency of steam methane reforming process the reaction space of the reformer
is often activated by nickel catalysts [55–59] and other catalysts [60–64].

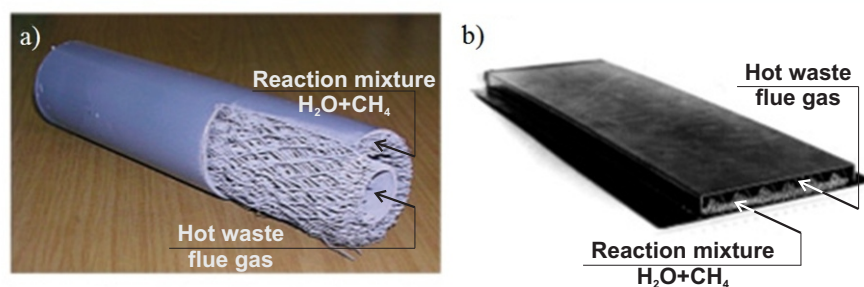


Figure 2: Reformer for thermochemical recuperation: a) tubular, b) plate

Fig. 3 shows the scheme of transformation of chemical energy into heat for

conventional furnaces without heat recovery (a) and for furnaces with TCR (b).

Analyzing the scheme in Fig. 1, it can be argued that if in the conventional fuel-consuming furnaces the conversion of chemical energy of fuel into heat occurs only by fuel combustion, then in the industrial furnaces with thermochemical recuperation the conversion of chemical energy of original fuel is divided into two stages. The first stage is the increasing of the low heating value of original methane by transformation of waste flue gas heat (H_{loss}) into chemical energy of the new synthetic fuel. The second stage is combustion of the new synthetic fuel that has low heating value greater than the low heating value of the original methane.

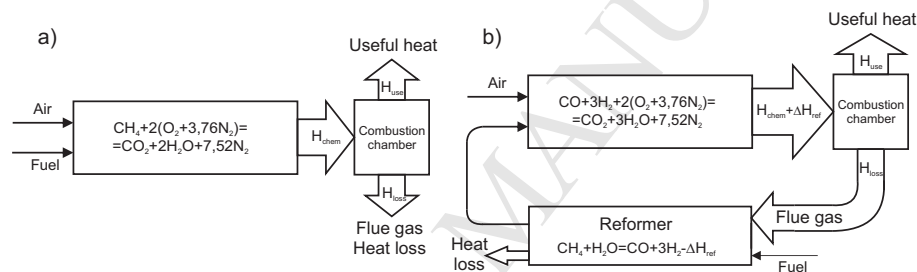


Figure 3: The scheme of transformation of chemical energy of fuel into heat for conventional furnaces (a) and for furnaces with TCR (b)

2.2. Chemical reactions

The chemical reaction of steam methane reforming occurs in the reformer presented in Fig.2. The overall reaction in steam methane reforming process to form new synthetic fuel which mostly consist of hydrogen (H_2) and carbon monoxide (CO) may include the different chemical equations listed in Table 1 [65–67]. On the other hand, according to thermodynamic analysis of the previous investigations [66, 67], only the first three equations showed in Table 1 occur significantly in steam methane reforming process while the rates of the other reactions (reactions (4)-(11)) are inappreciable. Therefore, the overall steam methane reforming process is described by reactions (1)–(3).

Table 1: Possible chemical reactions in the reformer [66, 67]

No.	Reaction	ΔH_{298} , (kJ/mol)
The overall reaction of SMR		
1	$\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$	206.1
2	$\text{CH}_4 + 2\text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + 4\text{H}_2$	165.0
Water gas shift reaction		
3	$\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$	-41.1
Dry methane reforming		
4	$\text{CH}_4 + \text{CO}_2 \rightleftharpoons 2\text{CO} + 2\text{H}_2$	247.3
5	$\text{CH}_4 + 3\text{CO}_2 \rightleftharpoons 4\text{CO} + 2\text{H}_2\text{O}$	330.0
Carbon formation		
6	$\text{CH}_4 \rightleftharpoons \text{C} + 2\text{H}_2$	74.8
7	$2\text{CO} \rightleftharpoons \text{C} + \text{CO}_2$	-173.3
8	$\text{H}_2 + \text{CO} \rightleftharpoons \text{C} + \text{H}_2\text{O}$	-131.1
9	$\text{CO}_2 + 2\text{H}_2 \rightleftharpoons \text{C} + 2\text{H}_2\text{O}$	-90.1
10	$\text{CH}_4 + 2\text{CO} \rightleftharpoons 3\text{C} + 2\text{H}_2\text{O}$	-187.6
11	$\text{CH}_4 + 2\text{CO}_2 \rightleftharpoons 2\text{C} + 2\text{H}_2\text{O}$	-15.3

Evidently, the efficiency of use of thermochemical waste-heat recuperation depends heavily on rate of methane conversion (reactions (1)-(2)), because it indicates how much of flue gas heat was transformed into chemical energy of new synthetic fuel. Thermodynamic equilibrium analysis of steam methane reforming allows to determine influence of feed composition, pressure and temperature on conversion rate of methane and enthalpy of overall steam methane reforming process.

Coke formation is another circumstance that needs to be investigated especially for high temperature levels. However, there is enough publications in literature that formation of solid carbon during steam methane reforming can be controlled by improving the coking resistance of the different catalysts as well as by regulating the operating conditions such as steam-to-methane ratio [68, 69].

In present research, steam-to-methane ratio in all investigated composition of inlet gas mixtures is either equal to or higher than unity, which shows that in the reformer there is enough steam to oxidize methane and prevent formation and deposition of coke [70, 71]. Therefore, coke formation can be ignored in this thermodynamic study.

3. Thermodynamic analysis

3.1. Methodology

Fully understanding TCR energy efficiency requires of application of the first law of thermodynamics (energy conservation law). By the first law, the amount of energy entering to any reformer at steady state must be equal to the amount of the reformer output [28, 72]. Put it another way, the quantity of hot flue gases heat that is converted to the chemical energy of a new synthetic fuel is determined by the enthalpy of the steam methane reforming reaction [29, 65]. On the other hand, the enthalpy of the steam methane reforming reaction is determined by the value of methane conversion rate [73–76].

The overall conversion rate of methane in the steam methane reforming process at different operation conditions (inlet steam-to-methane ratio and operation temperature) can be determined based on thermodynamic analysis. The influence of pressure, different inlet steam-to-methane ratio and the operating temperatures on the performances of SMR process were studied by means of total Gibbs free energy minimization via IVTANTHERMO [77–79].

The method of thermodynamic analysis of different chemical process by minimization of Gibbs free energy was introduced in author's previous publications [29] and it was also described in detail by other authors [80–84]. Thermodynamic analysis of SMR process via total Gibbs free energy minimization method was introduced to determine equilibrium compositions and conversation rate of methane. The method of Gibbs free energy minimization is based on the principles that the analyzed chemical scheme is thermodynamically auspicious if total Gibbs free energy has lowest value and differential of Gibbs free energy is zero

for a presented operation conditions [74, 85]:

$$dG^t = 0 \quad (1)$$

230 where G^t is value of total Gibbs free energy.

At pressure below 10 bar and high temperature, the reaction mixture can be considered as ideal [86]. Moreover, the synthesis gas composition from the reformer is determined, based on the following assumption: steam methane reforming is steady flow process with negligible change of kinetic and potential 235 energies; perfect insulation of reformer's wall with negligible heat loss to environment; chemical reactions with known reactants and products elementary composition are equilibrium.

Total Gibbs free energy of a SMR system is calculated by the sum of the chemical potential of all chemical components:

$$G^t = \sum n_i \mu_i \quad (2)$$

240 where n_i is number of moles of element i ; μ_i is chemical potential of element i .

The chemical potential of each element is presented as follow:

$$\mu_i = \Delta G_{fi}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (3)$$

where G_{fi}^0 is standard Gibbs free energy of formation of element i ; R is molar gas constant; T is temperature of the chemical system; f_i is fugacity of pure elements i ; f_i^0 is fugacity of pure elements i at the standard state.

245 If merge the expressions above, the equation for total Gibbs free energy of the chemical system can be written as:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (4)$$

The thermodynamic equilibrium analysis employing the Gibbs free energy minimization technique were performed by IVTANTHERMO [77–79], which is a process simulator for investigation of steady-state and unsteady-state process 250 of chemical systems developed by Joint Institute for High Temperatures (JIHT) of Russian Academy of Sciences. The software is able to model a single phase

or multiphase of multi-component in equilibrium condition. IVTANTHERMO includes an extensive database of thermodynamic properties of individual substances, programs for the database handling and a program, which allows the determination of thermodynamic properties each elements and equilibrium composition of the system to be examined.

Gibbs reactor in IVTHANTHERMO software is used to determine composition of products in reaction system under the conditions of minimization of Gibbs free energy. This reactor calculates the distribution of reaction products by minimizing the Gibbs free energy of each of the existing elements in the reaction system regardless of the reaction network. This software recommends Peng-Robinson equation of state due to its largest applicability range in terms of temperature and pressure. In addition; Peng- Robinson equation of state is regarded to be suitable to handle systems containing methane, steam, carbon monoxide and hydrogen [87]. Peng-Robinson equation of state was used in fugacity calculations [88]. To perform the thermodynamic analysis, thermodynamic properties of each species were determined at specified temperature and pressure and then, minimization was calculated to obtain equilibrium product and reactant compositions. Also some results were verified by Aspen-HYSYS version 8.4, which is a powerful program for modeling chemical systems [89–93].

3.2. Results of thermodynamic analysis

For further energy analysis, it is necessary to know the composition of the new synthetic fuel after reformer and the conversion rate of methane, depending on operating temperature, pressure and composition. The major gas components involved in the steam methane reforming process are hydrogen (H_2), carbon monoxide (CO), steam (H_2O), methane (CH_4) [65, 66]. Steam methane reforming process are also accompanied by slightly exothermic water-gas shift reaction (WGS), which is favored at relatively low temperatures below 1000K, and reverse water-gas shift-reaction, that is slightly endothermic and favored at temperatures above 1000 K [74].

The equilibrium conversion rates of methane are determined as:

$$X_{CH_4} (\%) = \frac{[CH_4]_{in} - [CH_4]_{out}}{[CH_4]_{in}} \times 100\% \quad (5)$$

The results of calculation of the conversion rates of steam, as well as hydrogen and carbon monoxide yield in this study are not presented, because for energy analysis it is necessary to know only the methane conversion rate and composition of reformat gas.

In the present paper, equilibrium conversion rate of methane and syn-gas composition were investigated as a function of operating temperature (600-1300 K) for different H_2O/CH_4 ratio at a fixed pressure of 10 bar. Obviously, for a more detailed thermodynamic analysis, a wider range of pressures should be investigated. Some results for other pressures are shown below.

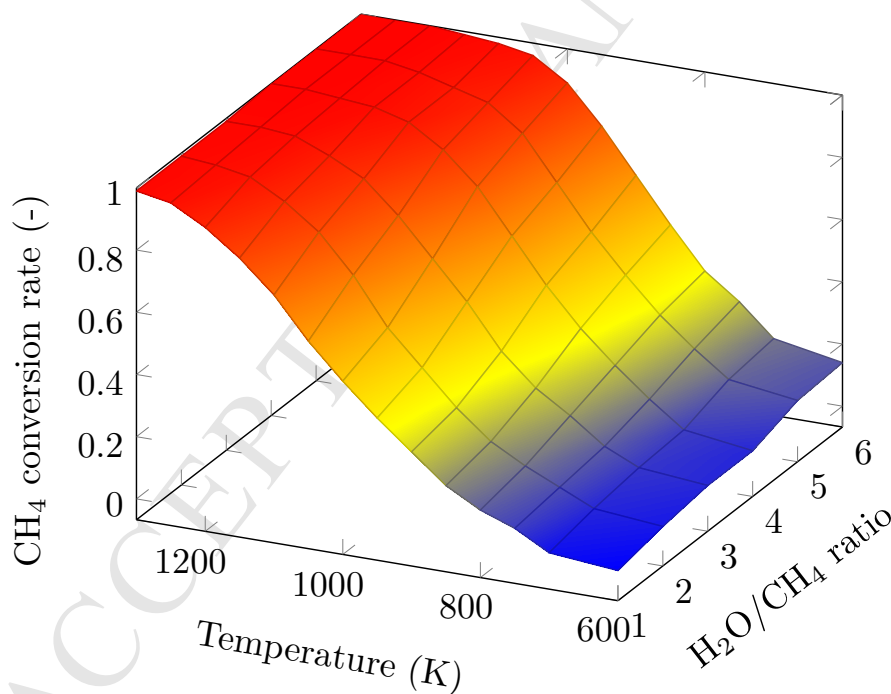


Figure 4: Effect of temperature on equilibrium conversion rate of methane for different inlet steam-to-methane ratio at pressure $p=10\text{bar}$

Fig. 4 presents equilibrium methane conversion rate as a function of steam-

to-methane ratio for different temperatures at a fixed pressure 10 bar. According to Fig. 4 steam additions have a noteworthy effect of methane conversion rate. The steam addition to reaction gas mixture at reformer inlet increases methane conversion rate massively; especially at operating temperatures below 1000 K, that is confirmed by Hegarty's article [94] and Ozkara-Aydinoglu's work [74]. With increasing temperature above 1200 K, about 100% conversion of methane is received. But on the other side, the steam addition to feed composition leads to increasing of non-combustible components in synthesis gas. This, in turn, reduces the efficiency of thermochemical recuperation.

Fig. 5 indicates synthesis gas composition for different temperatures at fixed steam-to-methane ration $\beta = 2$. The ratio $\beta = 2$ is considered because in this case a high methane conversion rate is achieved with a relatively low volume fraction of noncombustible components in the synthesis gas. Determination of the optimal ratio β is given below.

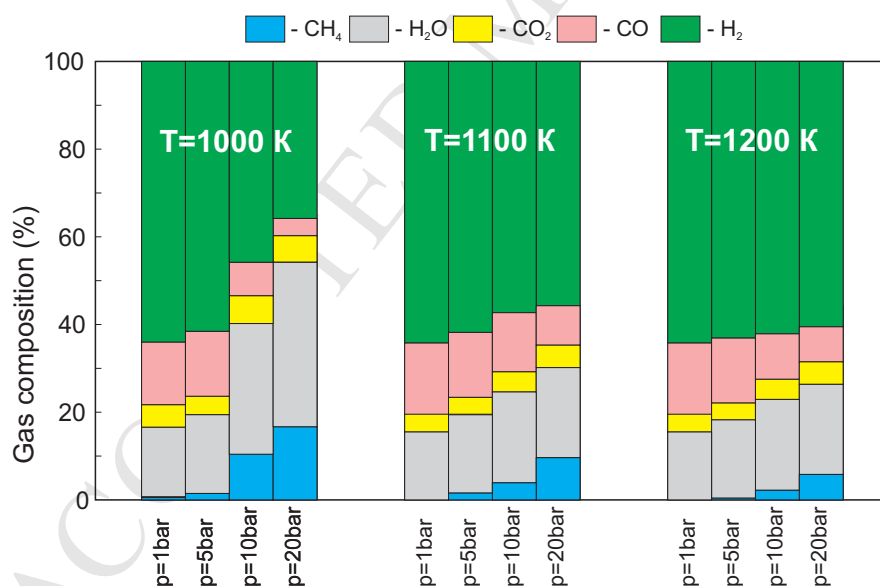


Figure 5: Effect of pressure on equilibrium conversion rate of methane for different temperatures at fixed steam-to-methane ratio $\beta = 2$

According to Fig. 5 the lowest volume concentration of non-combustible

components (H_2O and CO_2) in synthesis gas is observed for pressure below 10 bar at all investigated temperatures. The volume concentration of combustible components of syngas (hydrogen, carbon monoxide and unreacted methane) for 1200 K is from 81,5% to 72,8% for 1 bar and 20 bar, respectively. On the contrary, for 1000 K the volume concentration of combustible components for 1000 K at 20 bar is much lower - 58,4%.

The volume concentration of methane increased drastically with the pressure increasing for all analyzed feed composition at 1000 K, 1100 K and 1200 K. This trend indicates that decreasing pressure shifted steam reforming reaction in the direction of more methane consumption. This is a consequence of the Le Chatelier's principle. When steam methane reforming system at equilibrium is subjected to change in pressure, then the system readjusts itself to counteract (partially) the effect of the applied change and a new equilibrium is established. In other words, whenever a system in equilibrium is disturbed the system will adjust itself in such a way that the effect of the change will be nullified.

Analyzing the data shown in the Fig.5, it can be concluded that for the effective operation of thermochemical recuperation it is necessary to use a pressure of less than 10 bar, because in this case the volume concentration of non-combustible components in the syngas is minimal. However, steam is usually generated at pressure 5-10 bar. Also methane cracking reaction occurs in the reformer at pressure about 1 bar, which is poisoning the catalyst [56, 95]. Thereby, for the thermochemical recuperation, it can be recommended to use the pressure in the reformer (Fig. 1) of 5-10 bar.

The comparison of synthesis gas composition presented in Table 2. There are three comparable results. First is equilibrium syngas composition calculated via IVTANTHERMO. Second is equilibrium syngas composition obtained via ASPEN HYSYS. And third is the equilibrium composition of the syngas reported in Vekhivker and Kravchenko's work [30] for the methane steam reforming at steam-to-methane ratio $\beta = 2$. Correlation between the calculated results is obvious.

Table 2: Comparison of synthesis gas composition, % (mol/mol)

Temperature	Pressure	Source	Gas composition, %				
			H ₂	CO	H ₂ O	CO ₂	CH ₄
1000K	1 bar	IVTAN	63.01	14.32	16.32	5.23	1.12
		ASPEN	63.40	14.38	16.12	5.08	1.02
		Verkhivker[30]	63.42	14.43	16.1	5.12	0.92
1000 K	5 bar	IVTAN	60.66	14.82	17.32	5.55	1.65
		ASPEN	61.60	14.98	17.92	4.08	1.42
		Verkhivker[30]	61.44	14.93	17.97	4.18	1.48
1100K	1 bar	IVTAN	63.88	16.08	16.02	3.93	0.09
		ASPEN	63.65	16.14	16.15	3.98	0.08
		Verkhivker[30]	63.76	16.11	16.11	3.94	0.08
1100K	5 bar	IVTAN	61.28	15.02	17.87	4.24	1.59
		ASPEN	61.57	14.99	17.88	4.11	1.45
		Verkhivker[30]	61.45	14.93	17.96	4.18	1.48

4. Results and discussion

4.1. Reaction enthalpy

By the first law of thermodynamics, the amount of energy entering any reformer at steady state must be equal to the energy leaving that reformer [28, 72]. In other words, the quantity of hot flue gases heat that is transformed into the chemical energy of a new synthetic fuel is determined by the enthalpy of the steam methane reforming reaction [29, 65]. To simplify the energy analysis, the following assumptions were made. The main assumption is adiabatic system and negligible heat losses. The loss of pressure of gas flows in all elements of schematic diagram (Fig.1) is not taken into account.

Fig. 6 shows the variation of reaction enthalpy within temperature range of 600–1300 K at 10 bar pressure for different steam-to-methane inlet ratio ($\beta = 1...6$). It was observed that the total enthalpy of methane reforming reaction increased with the increase in temperature. The maximum reaction

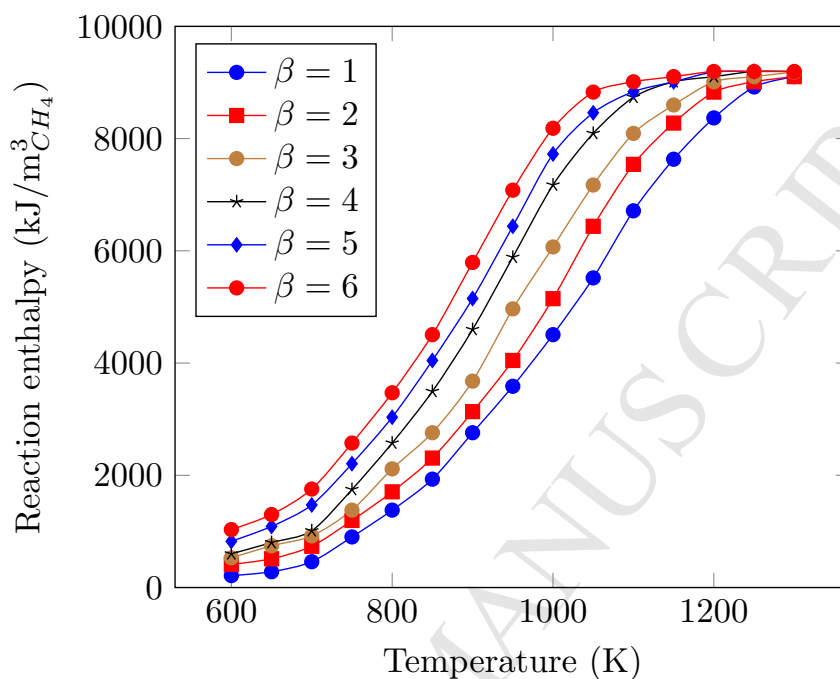


Figure 6: The variation of reaction enthalpy as a function of steam-to-methane ratio for various temperatures at 10 bar pressure

enthalpy is about 9200 kJ/m³ for all β ratios achieved at the temperatures of above 1200 K.

Evidently, the energy efficiency of using thermochemical waste-heat recuperation depends on the enthalpy of the steam methane reforming reaction. But on the other hand, an increase in the reaction enthalpy by the steam addition to the feed stock leads to increasing of non-combustible components in the synthesis gas and an excess demand energy to produce steam (according to scheme depicted in Fig. 1). The determination of the optimal steam-to-methane ratio for effective using of TCR is given below.

4.2. Transformation and recuperation rate

Fig. 7 shows the variation of the transformation coefficient as a function of steam-to-methane ratio for various temperatures at pressure 1 bar and 5 bar.

Here, transformation coefficient or transformation rate is the relation of low heating value of new synthesis fuel to heat of combustion of original methane.

365 Consequently, the total heat obtained at combustion of all the converted gases $\sum Q$ produced by 1 m³ methane reforming and the ratio of this heat to the heat, obtained by combustion of 1 m³ CH₄ is the transformation rate η_q .

$$\eta_q = \frac{\sum LHV_{sg}}{LHV_{CH_4}}, \quad (6)$$

where $\sum LHV_{sg}$ – low heating value of all synthesis gas produced by reforming of 1 mol methane [96, 97], kJ; LHV_{CH_4} – combustion heat of 1 mol of original methane [98], kJ ($LHV_{CH_4} = 802.34 \text{ kJ}$).

$$\sum LHV_{sg} = n \cdot (244.02 \cdot H_2 + 283.24 \cdot CO + 802.34 \cdot CH_4), \quad (7)$$

where n – number of synthesis gas mol, which is produced by reforming of 1 mol methane; H₂, CO, CH₄ – mole fraction of each components.

Fig. 7 depicts the increasing temperature of SMR process increases the combustion heat of the synthesis gas in comparison to the combustion heat of original methane. On the contrary, the increasing pressure reduces the conversion rate of steam methane reforming reaction as a consequence transformation rate is decreased. For 5 bar and 1000 K the maximum transformation coefficient is observed at about $\beta = 2$ and with increasing β increases slightly the transformation rate. At temperatures above 1100 K, the transformation coefficient increases sequentially with increasing ratio β .

Fig. 8 exhibit the effects of temperature on recuperation rate at pressure 10 bar for different steam-to-methane ratio β . The relation of recovered heat to waste flue gases heat is recuperation rate or recovery rate [99, 100]. In other words, recuperation rate is percentage of heat recovered and it can be calculated

385 as:

$$r = \frac{H_{SMR} + H_{sg}}{H_{fg}}, \quad (8)$$

where H_{SMR} – enthalpy of steam methane reforming reaction, kJ; H_{sg} – enthalpy of combustible components of synthesis gas, kJ; H_{fg} – enthalpy of hot flue gases after combustion chamber.

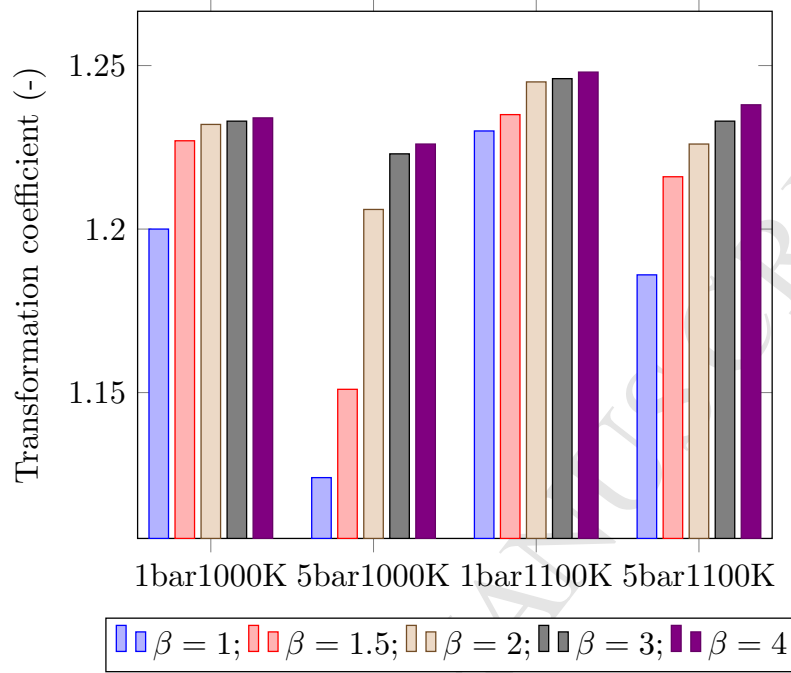


Figure 7: Variation of the transformation coefficient as a function of temperature for $\beta = 1$, 1.5, 2, 3 and 4 at different pressure

Here, recovered heat is calculated as sum of enthalpy of steam methane
 390 reforming reaction and heat of combustible components of synthesis gas. The
 enthalpy of incombustible components of syngas (H_2O and CO_2) is irrelevant
 in eq.(8), because this is ballast. Heat of combustible components of syngas is
 defined as follows:

$$H_{sg} = T_{sg} \sum (n_i \cdot C_{P_i}) - (H_w + H_{CH_4}), \quad (9)$$

where T_{sg} – synthesis gas temperature after reformer, K; n_i – numbers of moles
 395 of i combustible component (hydrogen, carbon monoxide, unreacted methane),
 mol; C_{P_i} – specific heat of i component, $kJ/(K \cdot mol)$; H_{steam} – enthalpy of steam
 after steam generator, kJ.

For analyzed schematic diagram the exhaust heat is utilized to evaporate
 water and produce steam for steam methane reforming. Otherwise, an addi-

400 tional energy is invested for this purpose. The enthalpy change of the reaction products occurs as a result of the use of the flue gas heat for steam production, and also by heating the reaction mixture in the reformer. This fact is taken into account in expression (9). The detailed heat flows of thermochemical waste-heat recuperation system are presented in Fig.10 and Fig.11.

405 All calculations are for 1 m³ of original methane.

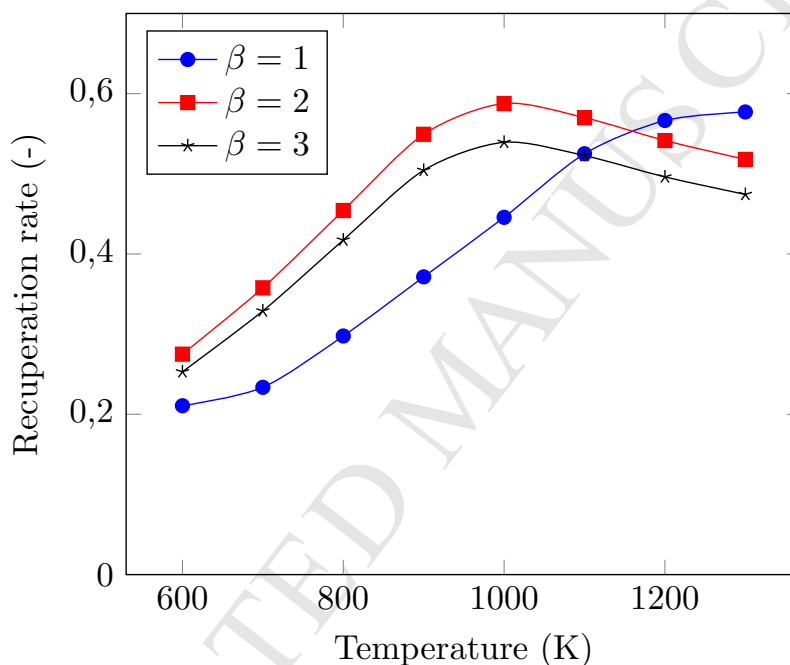


Figure 8: The variation of reaction enthalpy as a function of temperature for different steam-to-methane inlet ratio at 10 bar pressure

According to Fig.8 it is evident that the maximum recuperation rate is about $\beta = 2$ at about 1000 K. The recuperation rate is constantly increasing in the temperature range below 1000 K for $\beta=2$ and 3, because conversion rate (Fig. 4) is strongly increasing with increasing temperature. However, recuperation rate 410 decreases in temperature range above 1000 K for $\beta=2$ and 3, because an increasing of conversion rate slows down and a large amount of ballast steam is contained in the synthesis gas. On the contrary, for $\beta = 1$ the recuperation

rate is constantly increasing with increasing temperature due to small amount of ballast steam.

415 To understand the optimal values of the β ratio for different temperatures, Fig. 9 shows the dependence of the recuperation rate as a function of ratio β for different temperatures. The optimal β for temperature range below 1000 K is about 2, for temperature above 1200 K the optimal β ratio tends to unity.

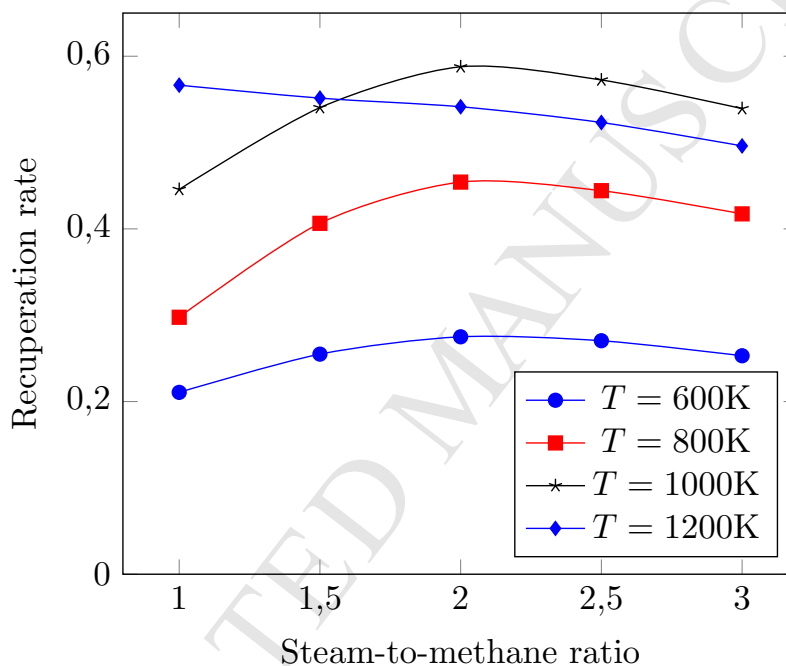


Figure 9: Recuperation rate as a function of ratio β for different temperatures

4.3. Heat balance

420 Heat balance contains the information regarding to the sources of heat inputs and outputs. A heat balance for thermochemical recuperation system contains the following flow of heat:

- input flows of heat balance: the heat of hot waste flue gases; the heat of inlet water and methane;

- 425 • output flows of heat balance: the heat for steam methane reforming reaction (reaction enthalpy), the heat for steam generation; the heat for syn.gas preheating; the heat of cold waste flue gas (exhaust gas)

Heat losses of gas transportation, wall losses and other heat losses are neglected.

430 The heat balance for system of thermochemical recuperation of waste flue gases heat are calculated for further conditions: original fuel is $\text{CH}_4=100\%$; temperatures of inlet water and methane are 27°C ; steam pressure is 10 bar and steam is saturated ($T_s = 453 \text{ K}$); excess air ratio $\lambda = 1$; temperature range of waste flue gas is 600-1300 K; temperature of synthesis gas equal as follows
 435 ($T_{fg} - 200$) K [101]. Calculation was performed using Mathcad. The results of the calculation are shown in Fig. 10 and in Fig. 11 for $\beta = 1$ and $\beta = 2$, respectively.

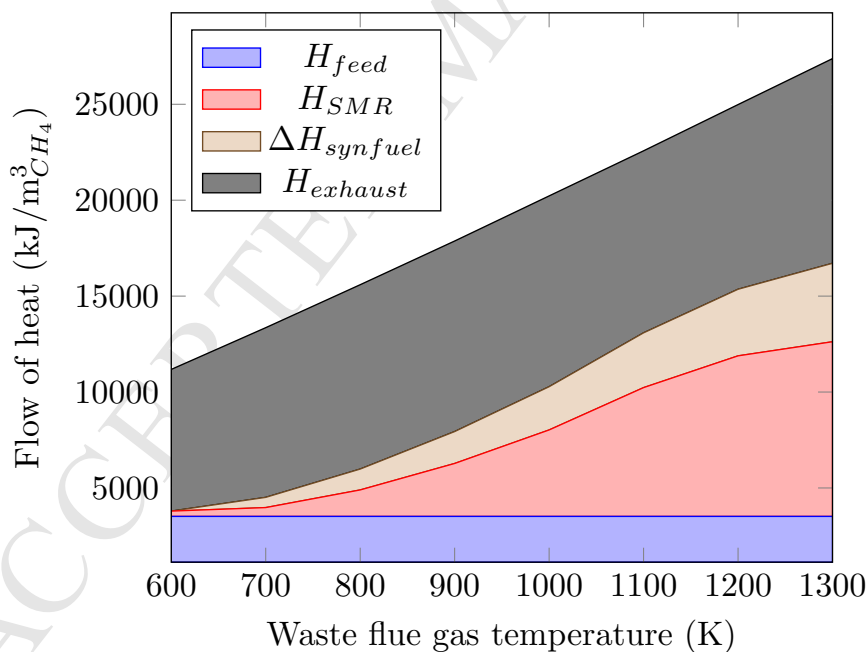


Figure 10: The heat balance of thermochemical waste-heat recuperation system for $\beta = 1$.

Here in Fig. 10 and Fig. 11 H_{feed} is the heat of reaction mixture in reformer

inlet at temperature $T=453$ K and pressure 10 bar and determined as follows:

$$H_{feed} = D \cdot h_g + M_{CH_4} \cdot C_{p_{CH_4}} \cdot T_{feed} - (H_w + H_{CH_4}), \quad (10)$$

where D – mass flow of steam, kg/s; h_g – specific enthalpy of saturated steam, kJ/kg; M_{CH_4} – mass flow of methane, kg/s; $C_{p_{CH_4}}$ – specific heat of methane, kJ/(K·kg).

The mass flow of steam in eq.(10) for $\beta = 1$ is 0,8036 kg for 1m^3 of original methane and 1,6072 kg for $\beta = 2$.

Also $\Delta H_{synfuel}$ is increasing enthalpy of syn.gas and determined as follows:

$$\Delta H_{synfuel} = H_{synfuel}^{out} - H_{feed}, \quad (11)$$

where $H_{synfuel}^{out}$ is total heat of syn.gas flow at outlet temperature, kJ.

Fig. 10 indicates that the minimum heat of exhaust gas is achieved in the temperature range of 1050–1200 K. This fact is confirmed by the curves shown in Fig. 8. Fig. 11 depicts that the minimum heat of exhaust gas is observed in the temperature range of 900–1050 K. The minimal heat of exhaust gases and maximum recuperation rate mean maximum energy efficiency of thermochemical waste-heat recuperation.

5. Conclusion

Based on the first law energy analysis of furnace with thermochemical recuperation the main conclusions drawn from the present research may be summarized as follows:

- The conversion rate of methane in SMR process depends on temperature, inlet steam-to-methane ratio and pressure. The steam additions have a notable effect methane conversion rate. The increasing of temperature from 600 to 1200 K leads to significant increasing of the conversion rate. With increasing the temperature to the level above 1200 K, 100% conversion of methane is obtained. The steam addition to feed stock at reformer inlet increases methane conversion rate scale up massively; especially at operating temperatures below 1000 K.

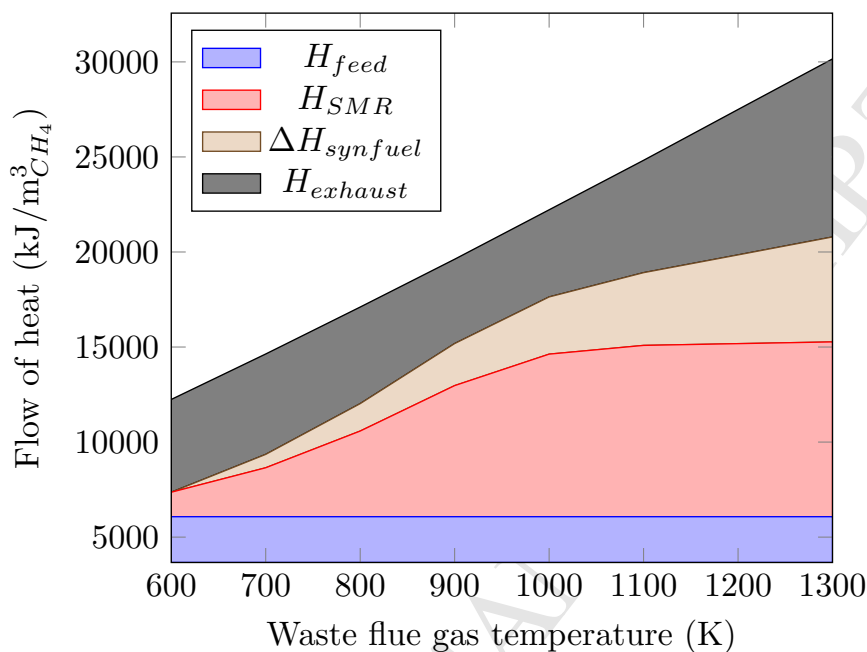


Figure 11: The heat balance of thermochemical waste-heat recuperation system for $\beta = 1$.

- The steam addition to feed composition leads to an increasing of non-combustible components in synthesis gas. This, in turn, reduces the efficiency of thermochemical recuperation.
- The lowest volume concentration of non-combustible components in synthesis gas is observed for pressure below 10 bar at temperature range 1000-1200 K at fixed steam-to-methane ratio $\beta = 1$. For the effective operation of thermochemical recuperation it is necessary to use a pressure of less than 10 bar, because in this case the volume concentration of non-combustible components in the syngas is minimal.
- The coefficient of heat transformation constantly increases with increasing of temperature and addition steam to feed stock. However, the transformation coefficient increases rapidly in ratio β range from 1 to 2 and increases slightly in β range above 2.

- The maximum recuperation rate is observed for temperature range below 1000 K at β ratio about 2, for temperature above 1200 K at β ratio tends to unity.

Summarizing the above, the optimal operation conditions for effective using of thermochemical recuperation of waste flue gases are: temperature range 900-1100 K for steam-to-methane inlet ratio $\beta = 2$ and temperature range above 1200 K for $\beta = 1$; optimal pressure is 5-10 bar.

The obtained results can be used for designing and optimizing of thermochemical heat recovery systems for industrial furnaces. Also, the results that obtained in this study can be used for a detailed analysis of the efficiency of a thermochemical reformer, a steam ejector and other elements of a thermochemical waste heat recuperation system.

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References

- [1] J. Gaede, J. Meadowcroft, A question of authenticity: Status quo bias and the international energy agency's world energy outlook, *Journal of Environmental Policy & Planning* 18 (5) (2016) 608–627.
- [2] S. Balitskiy, Y. Bilan, W. Strielkowski, D. Štreimikienė, Energy efficiency and natural gas consumption in the context of economic development in the european union, *Renewable and Sustainable Energy Reviews* 55 (2016) 156–168.
- [3] S. A. Solarin, I. Ozturk, The relationship between natural gas consumption and economic growth in opec members, *Renewable and Sustainable Energy Reviews* 58 (2016) 1348–1356.

- [4] D. E. Clark, C. Dybicz, A. Hanson, F. Nourzad, no shut-off policies and
 505 natural gas consumption, *Resource and Energy Economics* 48 (2017) 19–
 29.
- [5] A. Golyashev, V. Gimadi, A. C. Kurdin, A. Hanson, F. Nourzad, *Energy
 bulletin*, Analytical Center of the Government of Russia 28 (2016) 30.
- [6] A. N. Stulberg, Natural gas and the russia-ukraine crisis: Strategic re-
 510 straint and the emerging europe-urasia gas network, *Energy Research &
 Social Science*.
- [7] A. Makarov, V. Chupyatov, Energy efficiency and energy saving in russia,
International Journal of Global Energy (2000) (2016) 1–3.
- [8] C. E. S. Yearbook, Energy statistics division of national bureau of statis-
 515 tics (2011).
- [9] H.-k. LIAN, Y. Li, G.-y. SHU, C.-w. GU, An overview of domestic tech-
 nologies for waste heat utilization [j], *Energy Conservation Technology* 2
 (2011) 007.
- [10] E. Tian, Y.-L. He, W.-Q. Tao, Research on a new type waste heat recovery
 520 gravity heat pipe exchanger, *Applied Energy* 188 (2017) 586–594.
- [11] M. Wei, X. Zhao, L. Fu, S. Zhang, Performance study and application of
 new coal-fired boiler flue gas heat recovery system, *Applied Energy* 188
 (2017) 121–129.
- [12] C. Li, H. Wang, Power cycles for waste heat recovery from medium to high
 525 temperature flue gas sources—from a view of thermodynamic optimization,
Applied Energy 180 (2016) 707–721.
- [13] Q. Zhang, X. Zhao, H. Lu, T. Ni, Y. Li, Waste energy recovery and energy
 efficiency improvement in chinas iron and steel industry, *Applied Energy*
 191 (2017) 502–520.

- [14] S. Kang, H. Li, J. Lei, L. Liu, B. Cai, G. Zhang, A new utilization approach of the waste heat with mid-low temperature in the combined heating and power system integrating heat pump, *Applied Energy* 160 (2015) 185–193.
- [15] L. Miró, J. Gasia, L. F. Cabeza, Thermal energy storage (tes) for industrial waste heat (iwh) recovery: a review, *Applied Energy* 179 (2016) 284–301.
- [16] R. G. Beerkens, H. P. Muysenberg, Comparative study on energy-saving technologies for glass furnaces., *GLASTECH BER.* 65 (8) (1992) 216–224.
- [17] F. J. Nelson, Waste heat recovery from regenerative glass furnaces using an air extraction process, in: *51st Conference on Glass Problems: Ceramic Engineering and Science Proceedings, Volume 12, no. 3-4*, John Wiley & Sons, 2009, p. 568.
- [18] M. Barati, S. Esfahani, T. Utigard, Energy recovery from high temperature slags, *Energy* 36 (9) (2011) 5440–5449.
- [19] P. Li, Thermodynamic analysis of waste heat recovery of molten blast furnace slag, *International Journal of Hydrogen Energy*.
- [20] M. Mokhtar, S. Burns, D. Ross, I. Hunt, Exploring multi-objective trade-offs in the design space of a waste heat recovery system, *Applied Energy* 195 (2017) 114–124.
- [21] H. Xu, A. Romagnoli, J. Y. Sze, X. Py, Application of material assessment methodology in latent heat thermal energy storage for waste heat recovery, *Applied Energy* 187 (2017) 281–290.
- [22] Q. Yin, W.-J. Du, X.-L. Ji, L. Cheng, Optimization design and economic analyses of heat recovery exchangers on rotary kilns, *Applied Energy* 180 (2016) 743–756.
- [23] T. Zhu, K. Swaminathan-Gopalan, K. Stephani, E. Ertekin, Thermoelectric phonon glass electron crystal via ion beam patterning of silicon, arXiv preprint arXiv:1710.03375.

- [24] J. Chen, K. Li, C. Liu, M. Li, Y. Lv, L. Jia, S. Jiang, Enhanced efficiency of thermoelectric generator by optimizing mechanical and electrical structures, *Energies* 10 (9) (2017) 1329.
- 560 [25] J. He, T. M. Tritt, Advances in thermoelectric materials research: Looking back and moving forward, *Science* 357 (6358) (2017) eaak9997.
- [26] S.-J. Liang, B. Liu, W. Hu, K. Zhou, L. Ang, Thermionic energy conversion based on graphene van der waals heterostructures, *Scientific Reports* 7.
- 565 [27] G. P. Olmsted, JH, Heat engine efficiency enhancement-through chemical recovery of waste heat, 7th Intersociety Energy Conversion Engg. Con (1972) 241–248.
- [28] V. K. Chakravarthy, C. S. Daw, J. A. Pihl, J. C. Conklin, Study of the theoretical potential of thermochemical exhaust heat recuperation for internal combustion engines, *Energy & Fuels* 24 (3) (2010) 1529–1537.
- 570 [29] D. Pashchenko, Thermochemical recovery of heat contained in flue gases by means of bioethanol conversion, *Thermal Engineering* 60 (6) (2013) 438–443.
- [30] G. Verkhivker, V. Kravchenko, The use of chemical recuperation of heat in a power plant, *Energy* 29 (3) (2004) 379–388.
- 575 [31] K. Knoche, H. Richter, Reduction of combustion irreversibility in a gas turbine power plant through off-gas recycling.
- [32] O. Choi, M. C. Lee, Investigation into the combustion instability of synthetic natural gases using high speed flame images and their proper orthogonal decomposition, *International Journal of Hydrogen Energy* 41 (45) (2016) 20731–20743.
- 580 [33] S. Popov, I. Svistunov, A. Garyaev, E. Serikov, E. Temyrkanova, The use of thermochemical recuperation in an industrial plant, *Energy* 127 (2017) 44–51.

- [34] D. Rue, A. Kozlov, M. Khinkis, H. Kurek, Thermochemical recuperation to increase glass furnace energy efficiency, in: 74th Conference on Glass Problems, Wiley Online Library, 2014, pp. 81–92.
- [35] O. Cesana, M. Gutman, M. Shapiro, L. Tartakovsky, Internal combustion engine with thermochemical recuperation fed by ethanol steam reforming products-feasibility study, in: IOP Conference Series: Materials Science and Engineering, Vol. 147, IOP Publishing, 2016, p. 012109.
- [36] J. Rostrup-Nielsen, Steam reforming and chemical recuperation, *Catalysis Today* 145 (1) (2009) 72–75.
- [37] H. Kobayashi, Thermochemical regenerative heat recovery process, uS Patent 6,113,874 (Sep. 5 2000).
- [38] A. Poran, L. Tartakovsky, Performance and emissions of a direct injection internal combustion engine devised for joint operation with a high-pressure thermochemical recuperation system, *Energy* 124 (2017) 214–226.
- [39] F. D. Chuahy, S. L. Kokjohn, High efficiency dual-fuel combustion through thermochemical recovery and diesel reforming, *Applied Energy* 195 (2017) 503–522.
- [40] D. Pashchenko, Thermodynamic equilibrium analysis of combined dry and steam reforming of propane for thermochemical waste-heat recuperation, *International Journal of Hydrogen Energy* 42 (22) (2017) 14926–14935.
- [41] W. Cao, D. Zheng, Thermodynamic performance of o₂/co₂ gas turbine cycle with chemical recuperation by co₂-reforming of methane, *Fuel* 86 (17) (2007) 2864–2870.
- [42] J. Qin, S. long Zhang, W. Bao, W. li Yu, B. Yu, W. Zhou, D. Yu, Experimental study on chemical recuperation process of endothermic hydrocarbon fuel, *Fuel* 108 (2013) 445–450.

- [43] D. Cocco, V. Tola, G. Cau, Performance evaluation of chemically recuperated gas turbine (crgt) power plants fuelled by di-methyl-ether (dme), *Energy* 31 (10) (2006) 1446–1458.
- [44] N. Maruoka, T. Mizuochi, H. Purwanto, T. Akiyama, Feasibility study for recovering waste heat in the steelmaking industry using a chemical recuperator, *ISIJ international* 44 (2) (2004) 257–262.
- [45] M. Bogarra, J. Herreros, A. Tsolakis, A. York, P. Millington, Study of particulate matter and gaseous emissions in gasoline direct injection engine using on-board exhaust gas fuel reforming, *Applied Energy* 180 (2016) 245–255.
- [46] L. Zhu, Z. He, Z. Xu, X. Lu, J. Fang, W. Zhang, Z. Huang, In-cylinder thermochemical fuel reforming (tfr) in a spark-ignition natural gas engine, *Proceedings of the Combustion Institute* 36 (3) (2017) 3487–3497.
- [47] H. van Limpt, R. Beerkens, A. Habraken, Energy recovery from waste heat in the glass industry & thermo chemical recuperator, in: *73rd Conference on Glass Problems: Ceramic Engineering and Science Proceedings*, John Wiley & Sons, New York, USA, Vol. 34, 2013, p. 3.
- [48] Y. Y. Pechenegov, V. Simonov, B. Semenov, O. Y. Kosova, A. Mrakin, Oxidative pyrolysis of oil shale in tubular flow reactors with external heating, *Solid Fuel Chemistry* 51 (1) (2017) 40–43.
- [49] S. Sikirica, H. Kurek, A. Kozlov, M. Khinkis, Thermochemical recuperation improves furnace thermal efficiency, *Heat Treating Progress* 7 (5) (2007) 28.
- [50] D. Pashchenko, Comparative analysis of hydrogen/air combustion cfd-modeling for 3d and 2d computational domain of micro-cylindrical combustor, *International Journal of Hydrogen Energy*.

- [51] M. Kieffer, T. Brown, R. C. Brown, Flex fuel polygeneration: Integrating renewable natural gas into fischer–tropsch synthesis, *Applied Energy* 170 (2016) 208–218.
- 640 [52] Q. Yi, G.-s. Wu, M.-h. Gong, Y. Huang, J. Feng, Y.-h. Hao, W.-y. Li, A feasibility study for co₂ recycle assistance with coke oven gas to synthetic natural gas, *Applied Energy* 193 (2017) 149–161.
- [53] O. Omoregbe, H. T. Danh, C. Nguyen-Huy, H. Setiabudi, S. Abidin, Q. D. Truong, D.-V. N. Vo, Syngas production from methane dry reforming over
645 ni/sba-15 catalyst: Effect of operating parameters, *International Journal of Hydrogen Energy*.
- [54] S. K. Hong, S. K. Dong, J. O. Han, J. S. Lee, Y. C. Lee, Numerical study of effect of operating and design parameters for design of steam reforming reactor, *Energy* 61 (2013) 410–418.
- 650 [55] A. Antzara, E. Heracleous, A. A. Lemonidou, Energy efficient sorption enhanced-chemical looping methane reforming process for high-purity h₂ production: Experimental proof-of-concept, *Applied Energy* 180 (2016) 457–471.
- [56] R. K. Singha, A. Shukla, A. Yadav, S. Adak, Z. Iqbal, N. Siddiqui, R. Bal,
655 Energy efficient methane tri-reforming for synthesis gas production over highly coke resistant nanocrystalline ni–zro₂ catalyst, *Applied Energy* 178 (2016) 110–125.
- [57] M. Wu, E. Zhang, Q. Guo, Y. Wang, J. Qiao, K. Li, P. Pei, N/s-me (fe, co, ni) doped hierarchical porous carbons for fuel cell oxygen reduction reac-
660 tion with high catalytic activity and long-term stability, *Applied Energy* 175 (2016) 468–478.
- [58] B. Wang, J. Zhu, Z. Lin, A theoretical framework for multiphysics modeling of methane fueled solid oxide fuel cell and analysis of low steam methane reforming kinetics, *Applied Energy* 176 (2016) 1–11.

- 665 [59] C. Song, Q. Liu, N. Ji, Y. Kansha, A. Tsutsumi, Optimization of steam methane reforming coupled with pressure swing adsorption hydrogen production process by heat integration, *Applied Energy* 154 (2015) 392–401.
- [60] B. Zhang, L. Zhang, Z. Yang, Z. He, An experiment study of biomass steam gasification over nio/dolomite for hydrogen-rich gas production, 670 *International Journal of Hydrogen Energy*.
- [61] O. U. Osazuwa, H. D. Setiabudi, R. A. Rasid, C. K. Cheng, Syngas production via methane dry reforming: A novel application of smcoo 3 perovskite catalyst, *Journal of Natural Gas Science and Engineering* 37 (2017) 435–448.
- 675 [62] E. T. Kho, E. Lovell, R. J. Wong, J. Scott, R. Amal, Manipulating ceria-titania binary oxide features and their impact as nickel catalyst supports for low temperature steam reforming of methane, *Applied Catalysis A: General* 530 (2017) 111–124.
- [63] M. Zeppieri, P. Villa, N. Verdone, M. Scarsella, P. De Filippis, Kinetic 680 of methane steam reforming reaction over nickel-and rhodium-based catalysts, *Applied Catalysis A: General* 387 (1) (2010) 147–154.
- [64] W. Cassinelli, S. Damyanova, N. Parizotto, D. Zanchet, J. Bueno, C. Marques, Study of the properties of supported pd catalysts for steam and autothermal reforming of methane, *Applied Catalysis A: General* 475 (2014) 256–269. 685
- [65] D. Hoang, S. Chan, O. Ding, Kinetic and modelling study of methane steam reforming over sulfide nickel catalyst on a gamma alumina support, *Chemical Engineering Journal* 112 (1) (2005) 1–11.
- [66] J. Xu, G. F. Froment, Methane steam reforming, methanation and water-gas shift: I. intrinsic kinetics, *AIChE Journal* 35 (1) (1989) 88–96. 690

- [67] M. Soliman, A. Adris, A. Al-Ubaid, S. El-Nashaie, Intrinsic kinetics of nickel/calcium aluminate catalyst for methane steam reforming, *Journal of Chemical Technology and Biotechnology* 55 (2) (1992) 131–138.
- 695 [68] V. R. Choudhary, K. C. Mondal, Co₂ reforming of methane combined with steam reforming or partial oxidation of methane to syngas over Nd-CuO₃ perovskite-type mixed metal-oxide catalyst, *Applied Energy* 83 (9) (2006) 1024–1032.
- [69] D. L. Trimm, Coke formation and minimisation during steam reforming reactions, *Catalysis Today* 37 (3) (1997) 233–238.
- 700 [70] M. Yang, H. Papp, Co₂ reforming of methane to syngas over highly active and stable Pt/MgO catalysts, *Catalysis today* 115 (1) (2006) 199–204.
- [71] Ş. Özkara-Aydinoğlu, E. Özensoy, A. E. Aksoylu, The effect of impregnation strategy on methane dry reforming activity of Ce promoted Pt/ZrO₂, *International journal of hydrogen energy* 34 (24) (2009) 9711–9722.
- 705 [72] A. Gonzalez, E. Solorzano, S. Laux, U. Iyoha, K. Wu, H. Kobayashi, Operating experience with Optimelt regenerative thermo-chemical heat recovery for oxy-fuel fired glass furnaces, in: 76th Conference on Glass Problems: Ceramic Engineering and Science Proceedings, Volume 37, Issue 1 (Version A), no. 1, John Wiley & Sons, 2016, p. 17.
- 710 [73] H.-S. Wang, K.-Y. Huang, Y.-J. Huang, Y.-C. Su, F.-G. Tseng, A low-temperature partial-oxidation-methanol micro reformer with high fuel conversion rate and hydrogen production yield, *Applied Energy* 138 (2015) 21–30.
- 715 [74] Ş. Özkara-Aydinoğlu, Thermodynamic equilibrium analysis of combined carbon dioxide reforming with steam reforming of methane to synthesis gas, *International journal of hydrogen energy* 35 (23) (2010) 12821–12828.
- [75] K. Jabbour, N. El Hassan, A. Davidson, S. Casale, P. Massiani, Factors affecting the long-term stability of mesoporous nickel-based catalysts in

combined steam and dry reforming of methane, *Catalysis Science & Technology* 6 (12) (2016) 4616–4631.

[76] D. Czyłkowski, B. Hrycak, M. Jasiński, M. Dors, J. Mizeraczyk, Microwave plasma-based method of hydrogen production via combined steam reforming of methane, *Energy* 113 (2016) 653–661.

[77] L. Gurvich, V. Iorish, D. Chekhovskoi, V. Yungman, Ivtanthermo thermodynamic database and software system for the personal computer, NIST Special Database 5.

[78] G. V. Belov, V. S. Iorish, V. S. Yungman, Ivtanthermo for windows-database on thermodynamic properties and related software, *Calphad* 23 (2) (1999) 173–180.

[79] V. Iorish, G. Belov, et al., Ivtanthermo database-version 3.0, Glushko Thermocenter of Russian Academy of Sciences.

[80] G. Zeng, Y. Tian, Y. Li, Thermodynamic analysis of hydrogen production for fuel cell via oxidative steam reforming of propane, *International Journal of Hydrogen Energy* 35 (13) (2010) 6726–6737.

[81] S. Adhikari, S. Fernando, S. R. Gwaltney, S. F. To, R. M. Bricka, P. H. Steele, A. Haryanto, A thermodynamic analysis of hydrogen production by steam reforming of glycerol, *International Journal of Hydrogen Energy* 32 (14) (2007) 2875–2880.

[82] G. Nahar, S. Madhani, Thermodynamics of hydrogen production by the steam reforming of butanol: analysis of inorganic gases and light hydrocarbons, *International Journal of Hydrogen Energy* 35 (1) (2010) 98–109.

[83] K. Vasudeva, N. Mitra, P. Umasankar, S. Dhingra, Steam reforming of ethanol for hydrogen production: thermodynamic analysis, *International Journal of Hydrogen Energy* 21 (1) (1996) 13–18.

- [84] Y. Lwin, W. R. W. Daud, A. B. Mohamad, Z. Yaakob, Hydrogen production from steam-methanol reforming: thermodynamic analysis, *International Journal of Hydrogen Energy* 25 (1) (2000) 47–53.
- [85] S. Aktaş, M. Karakaya, A. K. Avcı, Thermodynamic analysis of steam assisted conversions of bio-oil components to synthesis gas, *International Journal of Hydrogen Energy* 34 (4) (2009) 1752–1759.
- [86] S. Chan, H. Wang, Thermodynamic analysis of natural-gas fuel processing for fuel cell applications, *International Journal of Hydrogen Energy* 25 (5) (2000) 441–449.
- [87] D.-Y. Peng, D. B. Robinson, A new two-constant equation of state, *Industrial & Engineering Chemistry Fundamentals* 15 (1) (1976) 59–64.
- [88] G. Halder, *Introduction to chemical engineering thermodynamics*, PHI Learning Pvt. Ltd., 2014.
- [89] L. E. Øi, Aspen hysys simulation of co₂ removal by amine absorption from a gas based power plant, in: *The 48th Scandinavian Conference on Simulation and Modeling (SIMS 2007)*; 30-31 October; 2007.
- [90] U. Zahid, F. N. Al Rowaili, M. K. Ayodeji, U. Ahmed, Simulation and parametric analysis of co₂ capture from natural gas using diglycolamine, *International Journal of Greenhouse Gas Control* 57 (2017) 42–51.
- [91] V. Feyzi, M. Beheshti, A. G. Kharaji, Exergy analysis: A co₂ removal plant using a-mdea as the solvent, *Energy* 118 (2017) 77–84.
- [92] T. N. Pham, N. V. D. Long, S. Lee, M. Lee, Enhancement of single mixed refrigerant natural gas liquefaction process through process knowledge inspired optimization and modification, *Applied Thermal Engineering* 110 (2017) 1230–1239.
- [93] E. F. Lazzaroni, M. Elsholkami, I. Arbiv, E. Martelli, A. Elkamel, M. Fowler, Energy infrastructure modeling for the oil sands industry: Current situation, *Applied Energy* 181 (2016) 435–445.

- [94] M. Hegarty, A. O'Connor, J. Ross, Syngas production from natural gas using zro 2-supported metals, *Catalysis Today* 42 (3) (1998) 225–232.
- 775 [95] Z. Wang, W. Fan, G. Zhang, S. Dong, Exergy analysis of methane cracking thermally coupled with chemical looping combustion for hydrogen production, *Applied Energy* 168 (2016) 1–12.
- 780 [96] A. Starik, A. Korobov, N. Titova, Combustion improvement in hcci engine operating on synthesis gas via addition of ozone or excited oxygen molecules to the charge: Modeling study, *International Journal of Hydrogen Energy*.
- [97] M. Mansourian, R. Kamali, Computational fluid dynamics analysis of a synthesis gas turbulent combustion in a round jet burner, *Acta Astronautica* 134 (2017) 133–140.
- 785 [98] M. A. Habib, S. S. Rashwan, M. A. Nemitallah, A. Abdelhafez, Stability maps of non-premixed methane flames in different oxidizing environments of a gas turbine model combustor, *Applied Energy* 189 (2017) 177–186.
- 790 [99] V. Vijay, A. Bhat, S. Shetty, N. Gurudatta, R. Sequeira, Design and fabrication of heat exchanger for waste heat recovery from exhaust gas of diesel engine, *Journal of Mechanical Engineering and Automation* 6 (5A) (2016) 131–137.
- [100] M. Diesel, S. Turbo, Waste heat recovery system (whrs) for reduction of fuel consumption, emissions and eedi.
- 795 [101] H. Kobayashi, K. Wu, R. Bell, Thermochemical regenerator: A high efficiency heat recovery system for oxy-fired glass furnaces, in: DGG/AcerS Conference, Aachen, Vol. 28, 2014.

Thermochemical waste-heat recuperation can be by steam methane reforming

Recuperation rate depends on steam-to-methane ratio, temperature and pressure

Results of thermodynamic analysis that performed by "IVTANTHERMO" are correlated with results that performed by Aspen HYSYS

The maximum recuperation rate is observed at steam-to-methane ratio of 2 for the temperature range of 900-1100K

The optimal pressure for using thermochemical recuperation in the reformer is 5-10bar